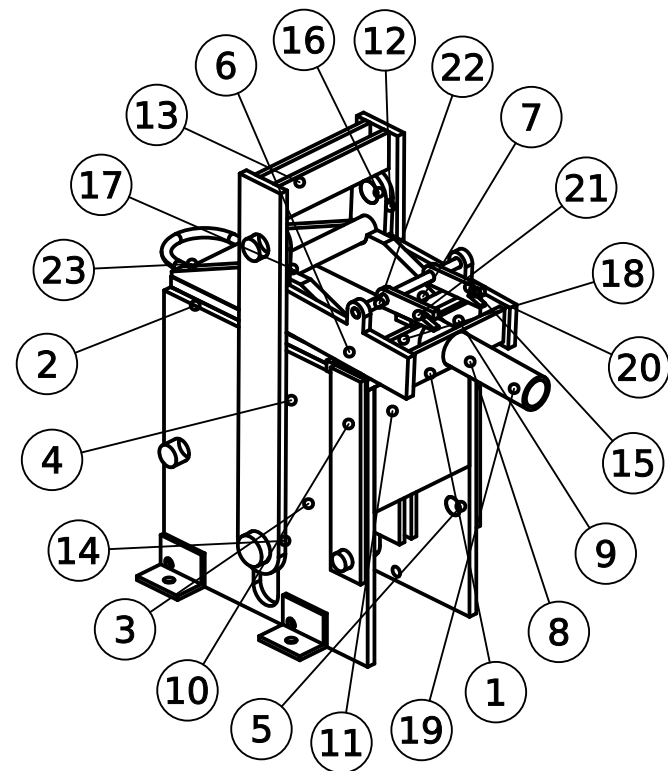
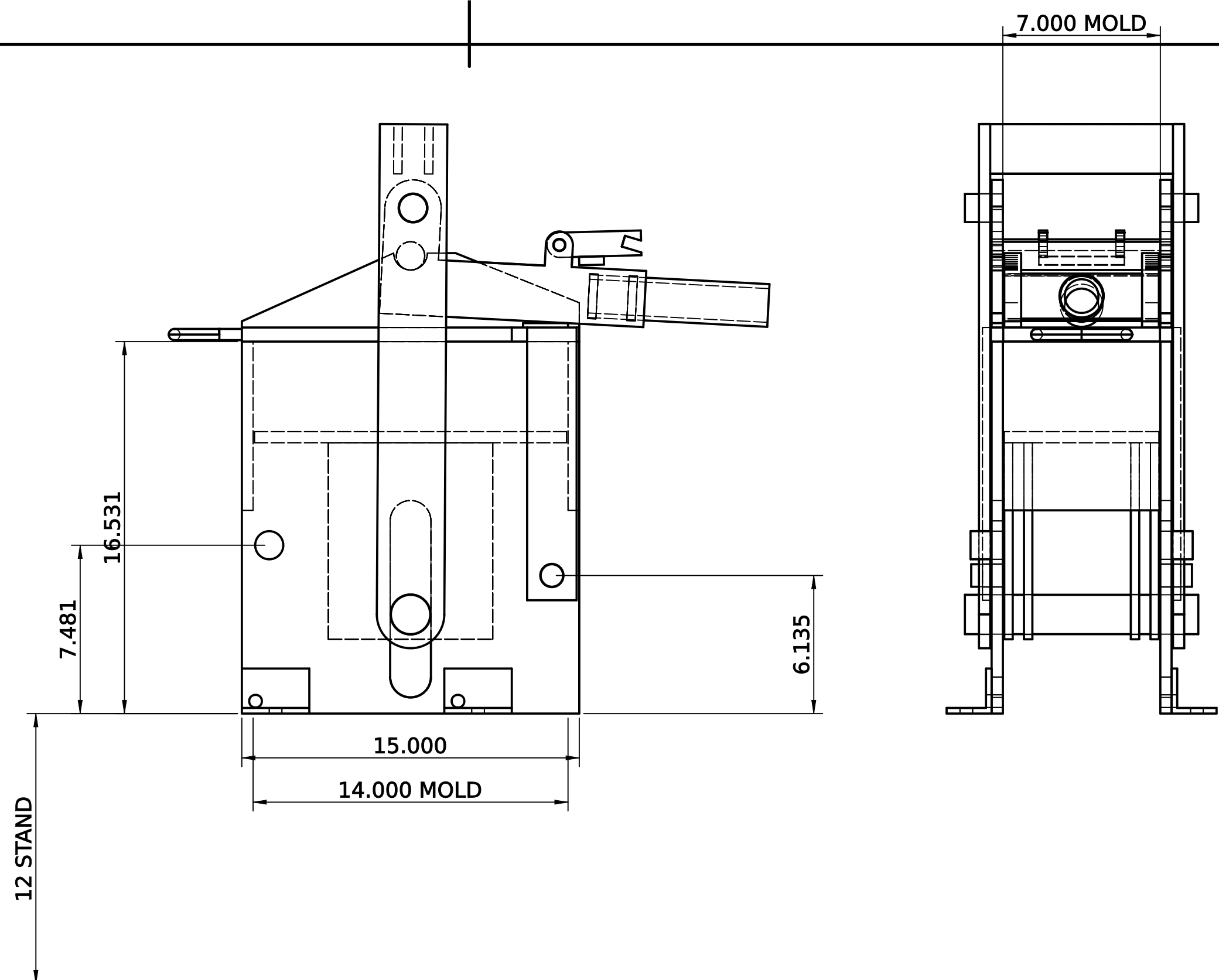




ISOMETRIC VIEW

- NOTES:
- 1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 - 2. BREAK SHARP EDGES, REMOVE BURRS.
 - 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 - 4. BRICK 14 X 7 X 4; FILL 6.75 DEEP (RATIO 1.69).
 - 5. SHOWN COMPRESSED AND LOCKED. HANDLE SHOWN BROKEN, FULL LENGTH 72.
 - 6. SEE SHEET 4 FOR OPERATION, LOADS AND STRENGTH CHECKS.
 - 7. STEEL WEIGHT ABOUT 230 LB.

23	1	Lid Handle	0.500" round 1018, bent	0.5	SCR-14x7x4-23
22	1	Latch Pivot	0.500" round 1018	0.4	SCR-14x7x4-22
21	1	Latch Bar	3/8 x 1-1/8 flat bar	0.4	SCR-14x7x4-21
20	2	Latch Hook	0.375" plate A36	0.4	SCR-14x7x4-20
19	1	Handle	1-1/2" sch 80 pipe	19.4	SCR-14x7x4-19
18	2	Handle Bridge	0.375" plate A36	1.2	SCR-14x7x4-18
17	1	Cam Pin	1.250" round 4140 prehard	2.8	SCR-14x7x4-17
16	2	Pin Q (stub)	1.250" round 4140 prehard	0.6	SCR-14x7x4-16
15	2	Head Cheek	0.500" plate A36	5.2	SCR-14x7x4-15
14	1	Main Pin P	1.750" round 4140 prehard	7.1	SCR-14x7x4-14
13	2	Cross Bar	0.375" plate A36	1.9	SCR-14x7x4-13
12	2	Yoke Arm	0.500" plate A36	9.2	SCR-14x7x4-12
11	4	Piston Web	0.375" plate A36	6.5	SCR-14x7x4-11
10	1	Piston Cap	0.500" plate A36	13.5	SCR-14x7x4-10
9	2	Handle Rest	0.195 x 1/2 flat bar	0.1	SCR-14x7x4-09
8	2	Lid Strap	0.375" plate A36	2.7	SCR-14x7x4-08
7	2	V Ramp	0.750" plate AR400	6.9	SCR-14x7x4-07
6	1	Lid Plate	0.625" plate A36	19.1	SCR-14x7x4-06
5	2	Lid Pivot Pin	1.000" round 1018	0.3	SCR-14x7x4-05
4	2	Fixed Pin	1.250" round 1018	0.5	SCR-14x7x4-04
3	4	Foot Clip	2 x 2 x 1/4 angle	0.8	SCR-14x7x4-03
2	2	End Plate	0.500" plate A36	7.4	SCR-14x7x4-02
1	2	Side Plate	0.500" plate A36	32.6	SCR-14x7x4-01
Item	Qty	Title	Stock	lb ea	Drawing no.
Material		Qty	Scale	Units	Projection
See parts list		1	1:10	inch	
Created by		Approved by	Document type		Status
J. Clements III		-	Assembly drawing		Draft
Legal owner		Title		Identification number	Sheet
J. Clements III		Simple CINVA-Ram Press		SCR-14x7x4-00	1/27
CEB Project		General assembly, 14x7x4 brick		Rev A	
				Date of issue	2026-09-30

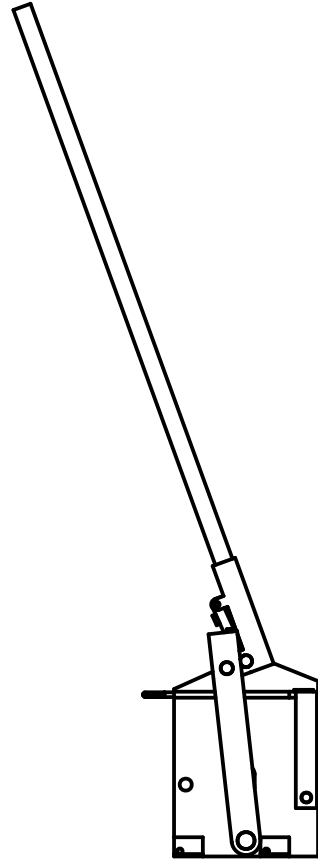


- NOTES:
- 1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 - 2. BREAK SHARP EDGES, REMOVE BURRS.
 - 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
 - 4. BRICK 14 X 7 X 4; FILL 6.75 DEEP (RATIO 1.69).
 - 5. SHOWN COMPRESSED AND LOCKED. HANDLE SHOWN BROKEN, FULL LENGTH 72.
 - 6. SEE SHEET 4 FOR OPERATION, LOADS AND STRENGTH CHECKS.

Material See parts list		Qty 1	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Assembly drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Simple CINVA-Ram Press Assembly views, 14x7x4 brick			Identification number SCR-14x7x4-00
		Rev A		Date of issue 2026-09-30	Sheet 2/27

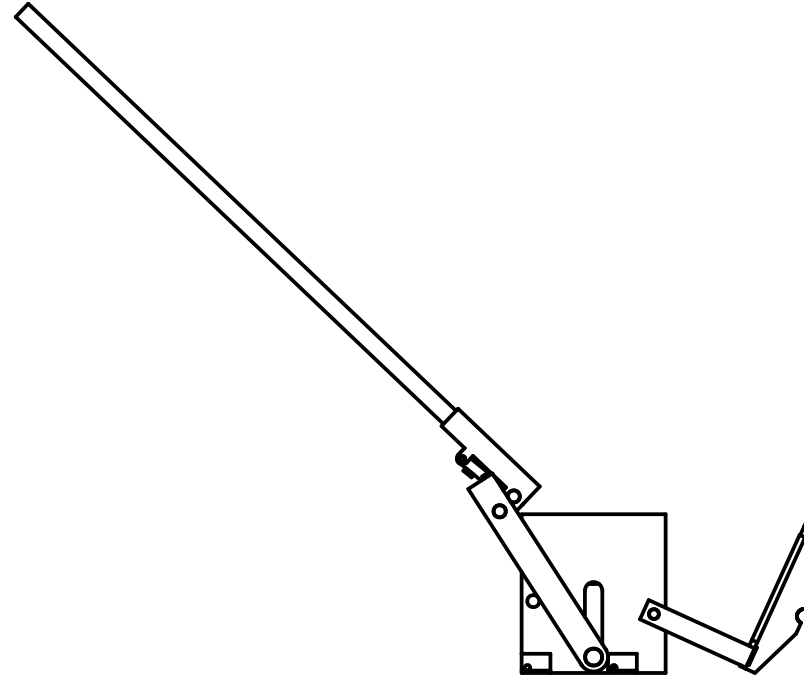
2 START OF COMPRESSION

yoke 6.4 deg, handle -110.0 deg, piston -6.750



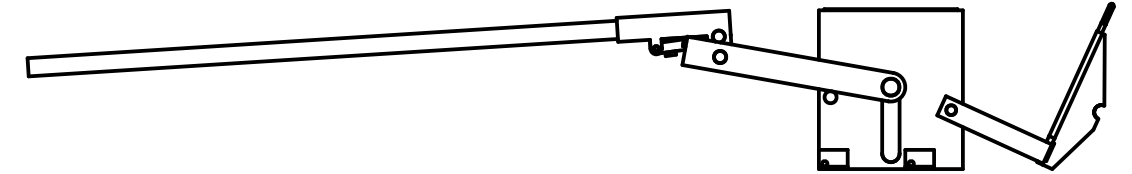
1 FILL

yoke 32.8 deg, handle -136.4 deg, piston -6.750



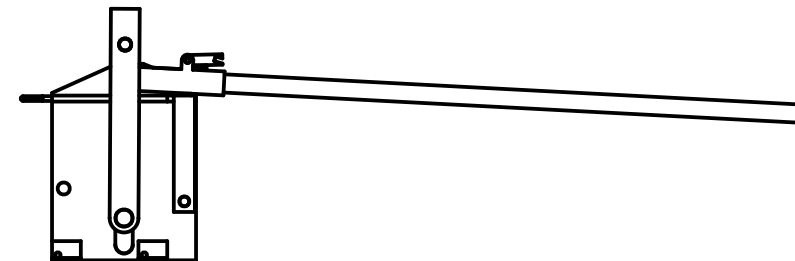
4 EJECT

yoke 80.0 deg, handle -183.6 deg, piston +0.125



3 COMPRESSED AND LOCKED

yoke -0.4 deg, handle 3.0 deg, piston -4.003



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. POSITIONS COMPUTED FROM THE LINKAGE; ALL FOUR AND THE MOTIONS BETWEEN THEM WERE CHECKED FOR CLEARANCE.
5. YOKE TILT FROM VERTICAL TOWARD THE FIXED PINS; HEAD ANGLE = DIRECTION CAM PIN -> Q FROM VERTICAL, POSITIVE TOWARD THE PULL END; PISTON = PISTON TOP RELATIVE TO THE MOLD TOP.

Material	-	Qty	1	Scale	1:20	Units	inch	Projection	
Created by	J. Clements III	Approved by	-	Document type	Assembly drawing	Status	Draft		
Legal owner	J. Clements III CEB Project	Title				Identification number		Sheet	
		Simple CINVA-Ram Press Operating positions, 14x7x4 brick				SCR-14x7x4-00		3/27	
		Rev	A	Date of issue		2026-09-30			

PRESS DATA

Brick (pressed)	14 x 7 x 4 in, 98 sq in face
Scale from his press	x 1.217 long, x 1.167 wide, x 1.102 tall
Loose fill depth	6.750 in (his 6-1/8); mold top 28.5 in above the ground on a 12 in stand
Working pressure	200 psi = 19,600 lbf on the piston
Design (overfill) load	306 psi = 29,988 lbf, all parts checked
Knee	cam arm 2.132, head -110 to +3 deg, yoke P-Q 18.063
Peak push on handle	123 / 163 / 204 lbf at 150 / 200 / 250 psi (72 in handle)
Ejection	piston rises 6.875 in; yoke tilts 33 to 80 deg; about 112 lbf on the handle
Steel weight	about 230 lb

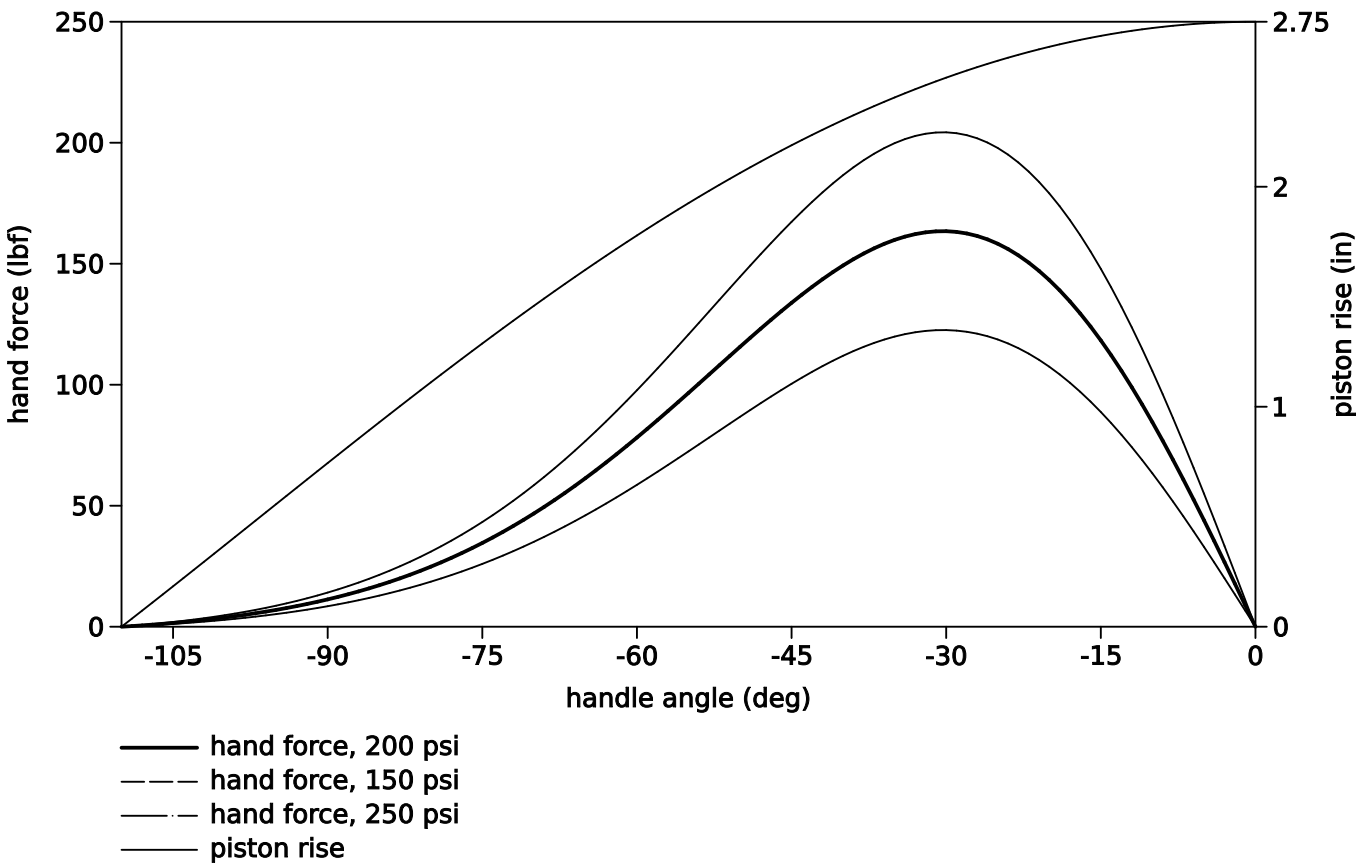
OPERATION

1. Start: latch shut, yoke leaning on the fixed pins, piston at the bottom, lid swung over onto the stand.
2. Oil the mold. Fill loose soil to the top, press it into the corners, strike off level. Swing the lid back on by its horseshoe handle.
3. Lift the handle and yoke upright together: the cam pin rides up the long leg of the ramps and drops into the scoops.
4. Lift the latch bar to free the head. Pull the handle over toward the lid pivots until the head cheeks land on the rests, 3 deg past centre: the brick is pressed and the handle stays down.
5. Raise the handle back past upright (the piston drops away from the brick), drop the latch over the cross bar, and tilt handle and yoke back onto the fixed pins.
6. Lift the lid by its horseshoe handle and swing it over onto the stand. Push the handle down: the piston lifts the brick clear. Lift it off by its ends.
7. Let the handle rise: the piston drops back to the bottom for the next fill.

STRENGTH CHECKS AT 29,988 LBF (ksi)

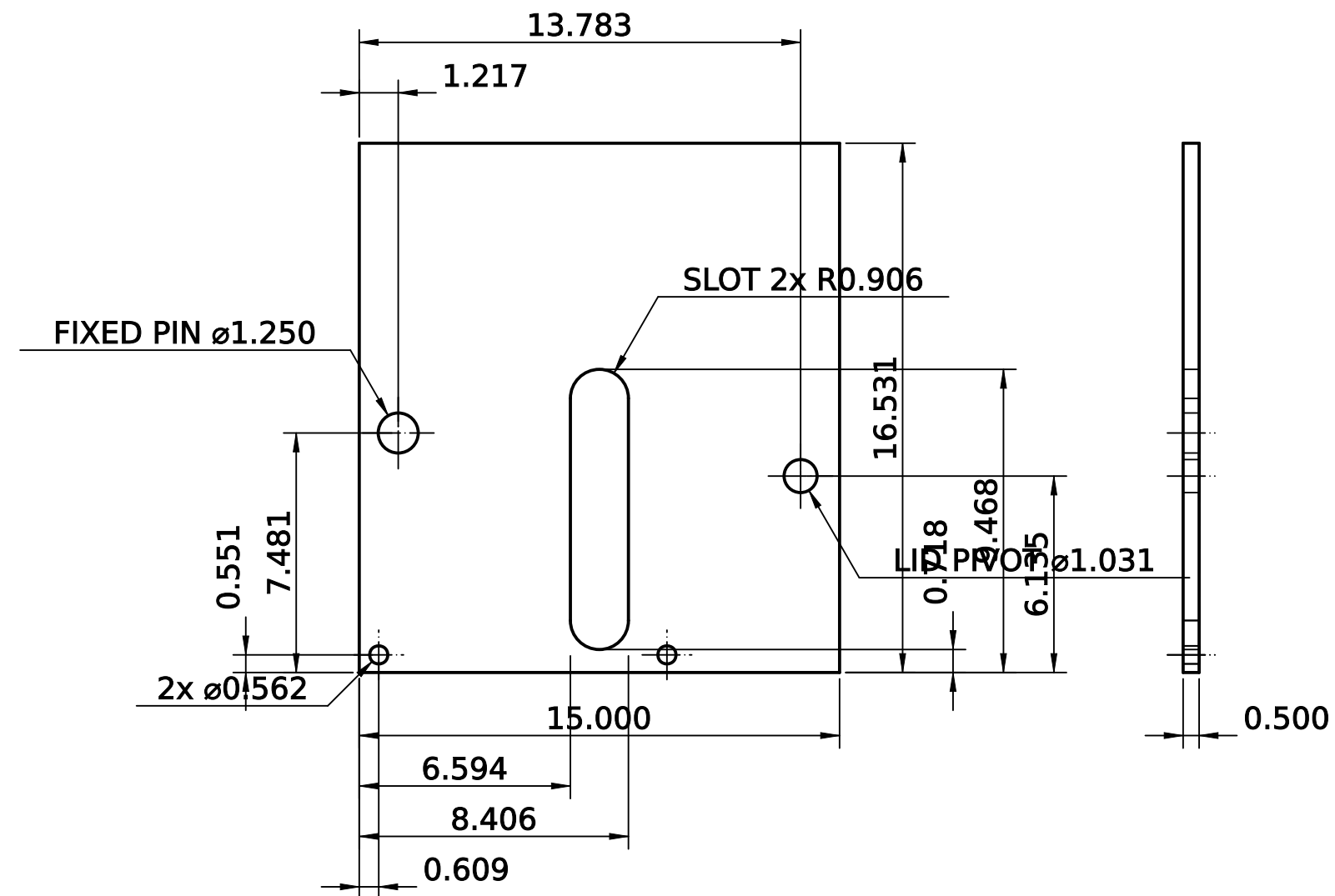
Item	Basis	Stress	Allow	Ratio
Main pin P, bending	M = 22,491 lbf in, 4 webs, dia 1.750 4140	42.7	57.0	0.75
Main pin P, shear	F/2 per side	6.2	38.0	0.16
Pin Q, bending	M = F/2 x 0.56, dia 1.250 4140	44.0	57.0	0.77
Cam pin, bending	M = F/2 x 0.69, dia 1.250 4140	53.8	57.0	0.94
Yoke arm, net tension at P	F/2 on 0.61 sq in	24.6	29.0	0.85
Yoke arm, pin bearing	F/2 on 1.75 x 0.50	17.1	32.4	0.53
Cheek, Q pin bearing	F/2 on 1.25 x 0.50	24.0	32.4	0.74
Piston web, pin bearing	F/4 on 1.75 x 0.375	11.4	32.4	0.35
Side plate, bending	K0 0.5, band 4 in, span 14, t 0.500	22.3	23.8	0.94
End plate, bending	K0 0.5, span 7 fixed ends, t 0.500	15.0	23.8	0.63
Lid and ramps, bending	M = F L / 8 at the scoop, l = 7.4 in^4	15.1	23.8	0.64
Lid plate between ramps	strip span 5.38, t 0.625	17.0	23.8	0.71
Piston cap, bending	strip span 4.75, t 0.500	20.7	23.8	0.87
Piston web, bending	F/4 over 7.30, supported at the pin	1.4	23.8	0.06
Cam pin in scoop, contact	R 0.625 in R 0.750, AR400	167.2	241.7	0.69
Fixed pin, bending (eject)	1,500 lbf per side x 1.19	9.3	33.0	0.28
Latch hooks, bearing (tilting)	2,684 lbf on 2 x 0.375 x 0.375	9.5	32.4	0.29

HAND FORCE AND PISTON RISE OVER THE STROKE



- NOTES:
- 1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 - 2. BREAK SHARP EDGES, REMOVE BURRS.
 - 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 - 4. FIRST-ORDER HAND CALCULATIONS; BUILD ONE PRESS AND LOAD-TEST BEFORE PRODUCTION.
 - 5. ALLOWABLES: A36 0.66 FY BENDING, 0.5 FU NET TENSION; 4140 PREHARD 0.6 FY; AR400 CONTACT 0.3 P $\leq$ 0.5 FY.
 - 6. SOIL MODEL: PRESSURE RISES AS $(E^5 X - 1)/(E^5 - 1)$ OVER THE STROKE; LATERAL WALL PRESSURE 0.5 X VERTICAL.
 - 7. EJECT HAND FORCE ASSUMES 1,000 LBF TO BREAK THE BRICK LOOSE; GREASE THE SCOOPS, PINS AND FIXED PINS.

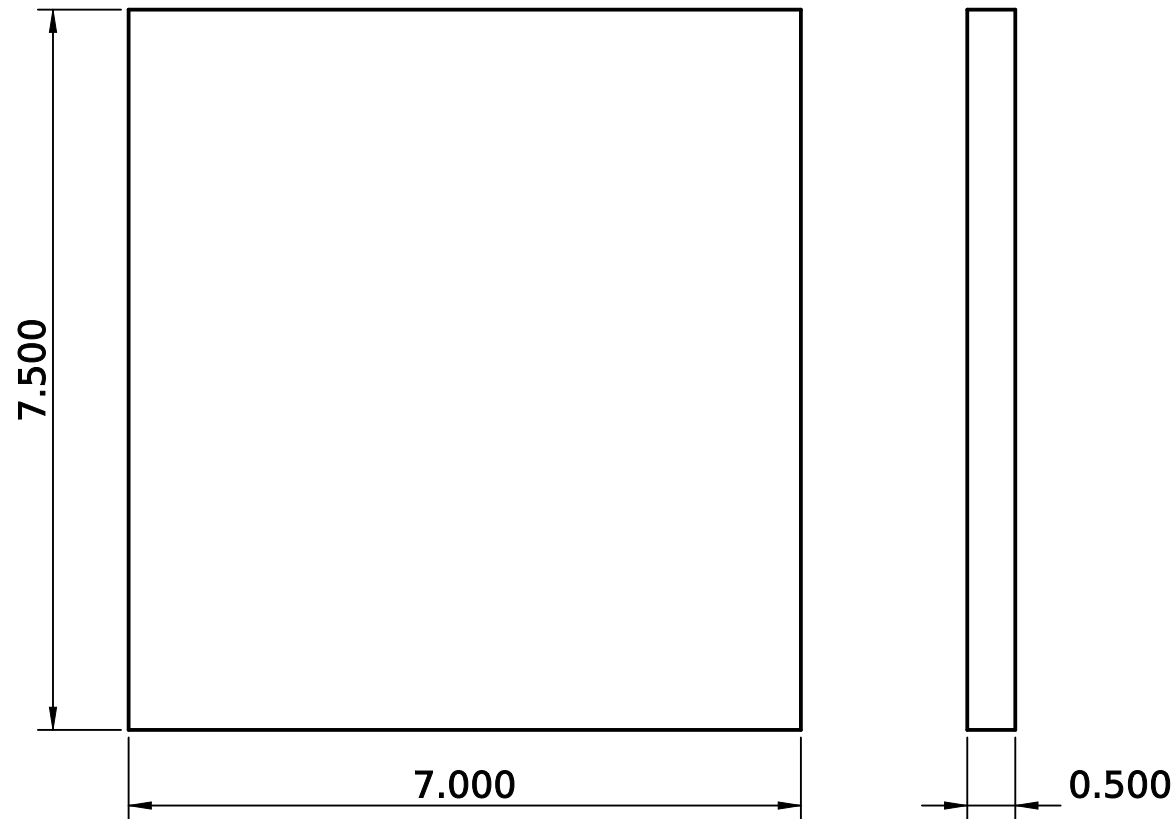
Material	Qty	Scale	Units	Projection
-	1	-	inch	
Created by	Approved by	Document type	Status	
J. Clements III	-	Design data	Draft	
Legal owner	Title	Identification number	Sheet	
J. Clements III	Simple CINVA-Ram Press	SCR-14x7x4-00	4/27	
CEB Project	Operation and design data, 14x7x4 brick	Rev A	Date of issue	2026-09-30



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. SLOT WIDTH IS HIS ERROR FIXED: HIS DRAWING SHOWS 2.000 FOR A 1 IN PIN; CUT IT PIN + $1/16$.
5. THE SLOT IS CENTRED ON THE PLATE LENGTH AND STOPS BELOW THE LOOSE SOIL AT FILL.
6. WELD THE END PLATES BETWEEN THE SIDE PLATES, OUTSIDE ONLY; NO WELD OR SPATTER INSIDE THE MOLD.

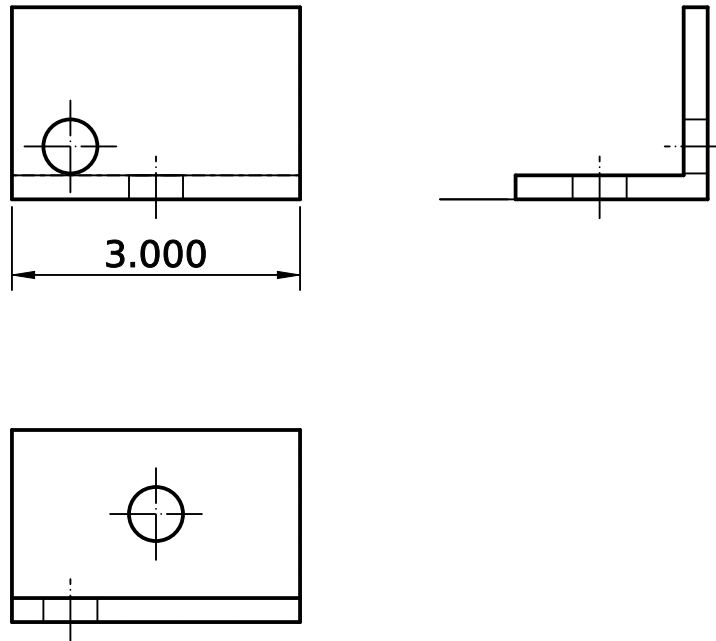
Material 0.500" plate A36		Qty 2	Scale 1:5	Units inch	Projection
Created by J. Clements III		Approved by -		Document type Detail drawing	
Legal owner J. Clements III CEB Project		Title Side Plate Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-01
		Rev A		Date of issue 2026-09-30	Sheet 5/27



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BETWEEN THE SIDE PLATES, TOP FLUSH WITH THEM. MOLD INSIDE MUST BE SQUARE: CHECK DIAGONALS.

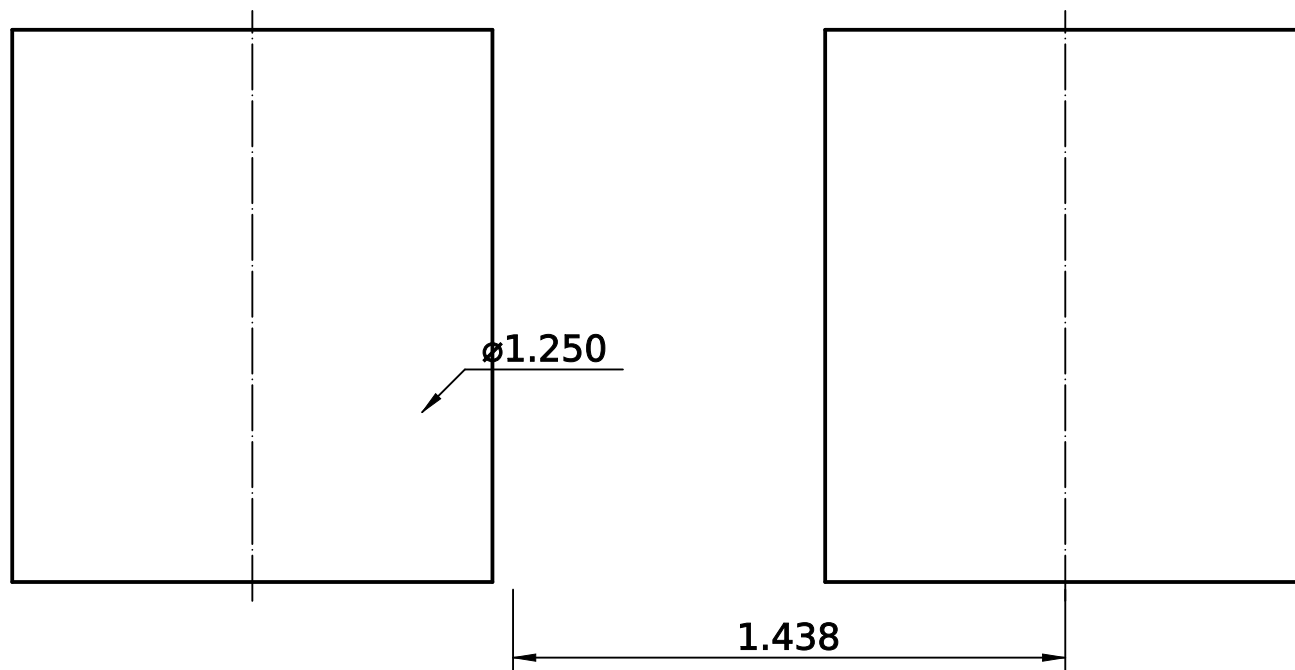
Material 0.500" plate A36		Qty 2	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title End Plate Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-02		Sheet 6/27
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BOLT THROUGH THE SIDE PLATE'S BOTTOM-CORNER HOLES ($1/2$ BOLTS), AND LAG OR BOLT TO THE STAND.
5. THE STAND TOP IS 12 IN ABOVE THE GROUND; ANY SOLID BENCH OR TIMBER FRAME WILL DO.

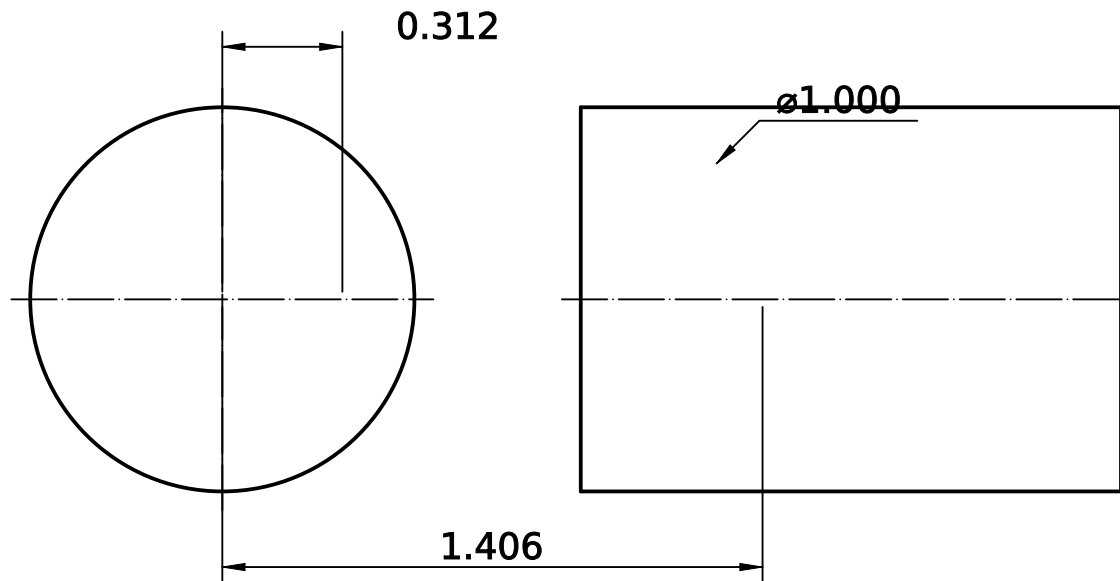
Material 2 x 2 x 1/4 angle		Qty 4	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Foot Clip Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-03
		Rev A		Date of issue 2026-09-30	Sheet 7/27



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH THE SIDE PLATE HOLE, INNER END FLUSH WITH THE INSIDE FACE; WELD OUTSIDE.
5. THE YOKE ARM LIES ON IT TO EJECT AND TO FILL. HEIGHT SET SO THE EJECT ENDS AT 80 DEG (HIS 7 IN X 1.102 = 7.71).

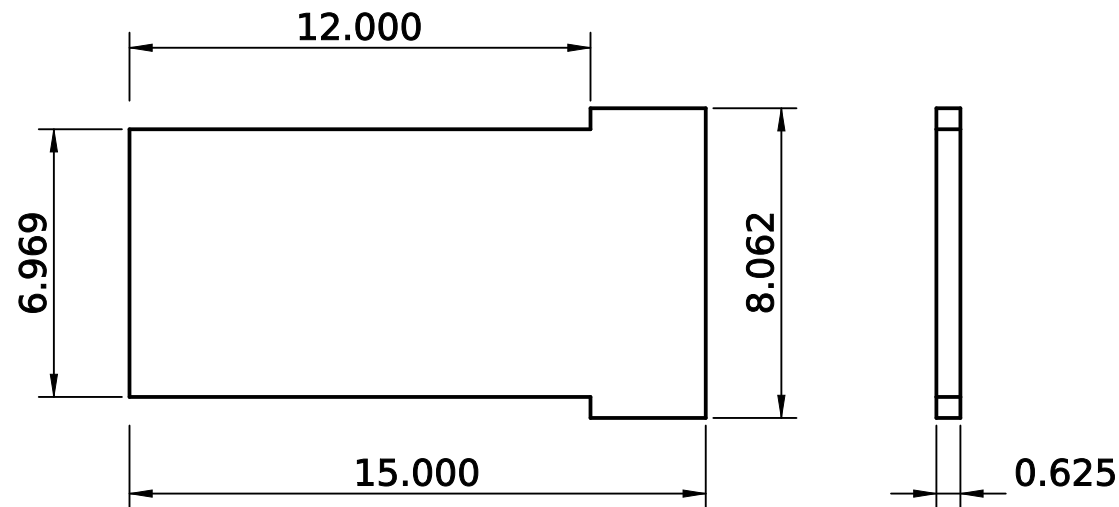
Material 1.250" round 1018		Qty 2	Scale 2:1	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Fixed Pin Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-04		Sheet 8/27
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. WELD IN THE SIDE PLATE, INNER END FLUSH INSIDE; STRAP ON THE OUTSIDE, WASHER AND 1/8 COTTER.

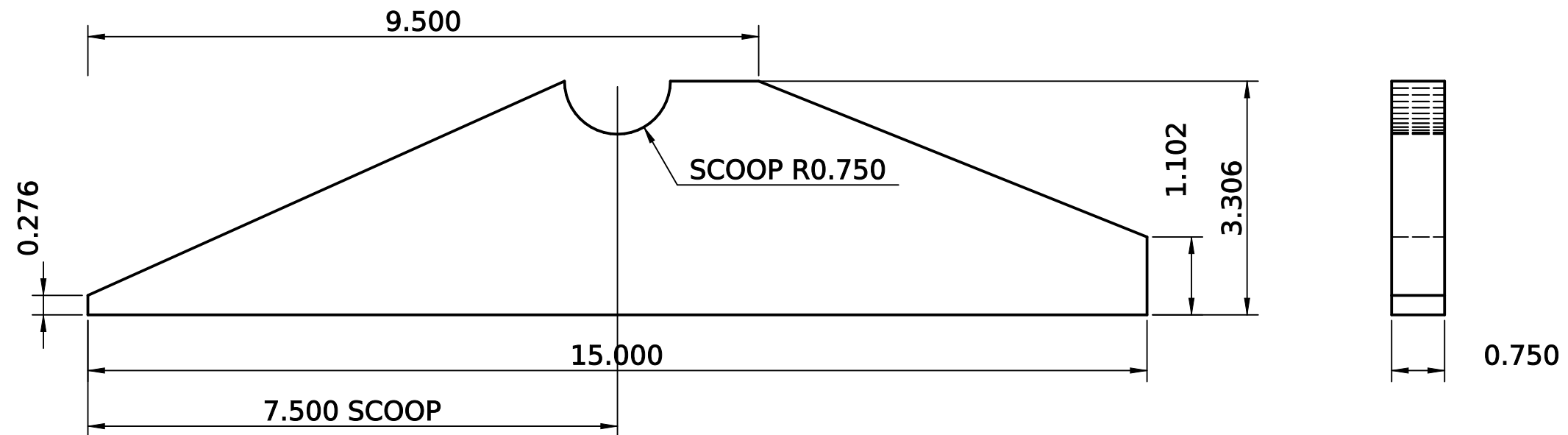
Material 1.000" round 1018	Qty 2	Scale 2:1	Units inch	Projection 
Created by J. Clements III	Approved by -	Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Lid Pivot Pin Simple CINVA-Ram Press, 14x7x4 brick	Identification number SCR-14x7x4-05	Sheet 9/27	
		Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. RESTS ON THE END PLATES. THE NARROW PART STAYS $1/64$ INSIDE THE SIDE PLATES' INNER FACES (THE HEAD SWINGS DOWN PAST IT); THE WIDE PART AT THE PULL END ALSO SITS ON THE SIDE PLATES.
5. UNDERSIDE FLAT AND CLEAN: IT IS THE BRICK'S TOP FACE.

Material 0.625" plate A36		Qty 1	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Lid Plate Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-06
		Rev A		Date of issue 2026-09-30	Sheet 10/27



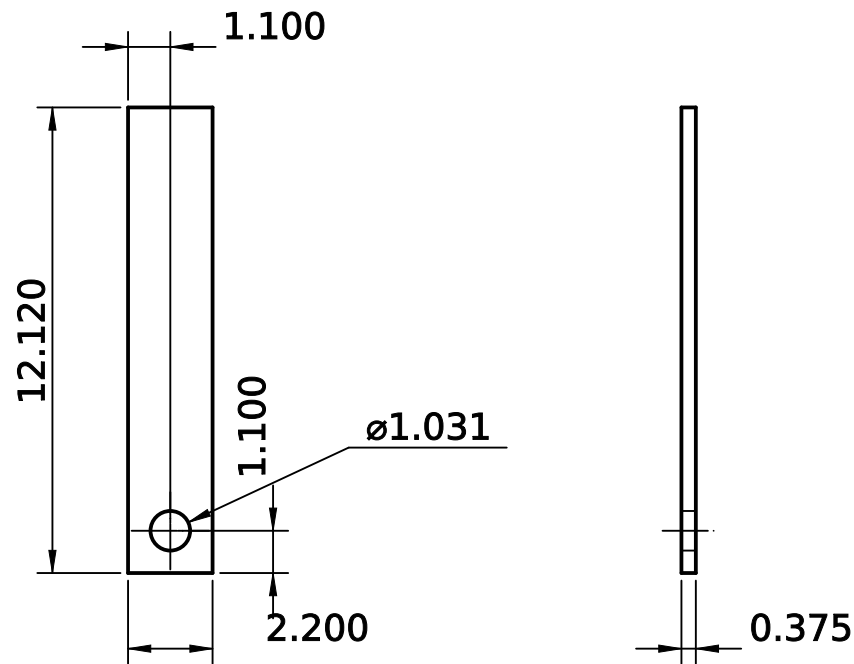
RAMP TOP ORDINATES (X FROM LID -X EDGE, H ABOVE RAMP BOTTOM); SCOOP PER DETAIL

X	H	X	H	X	H	X	H	X	H
0.000	0.276	3.500	1.847	7.000	2.749	10.500	2.905	14.000	1.503
0.500	0.500	4.000	2.071	7.500	2.556	11.000	2.705	14.500	1.302
1.000	0.724	4.500	2.296	8.000	2.749	11.500	2.505	15.000	1.102
1.500	0.949	5.000	2.520	8.500	3.306	12.000	2.304		
2.000	1.173	5.500	2.745	9.000	3.306	12.500	2.104		
2.500	1.398	6.000	2.969	9.500	3.306	13.000	1.903		
3.000	1.622	6.500	3.194	10.000	3.106	13.500	1.703		

NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. HIS V PLATE SCALED: LONG LEG TO THE -X END (FIXED-PIN END), HALF-ROUND SCOOP CENTRED ON THE MOLD, FLAT, SHORT LEG TO THE +X END.
5. SCOOP R0.750 (CAM PIN + 1/8, AS HIS R0.625 FOR A 1 IN PIN); DRESS IT SMOOTH.
6. WELD TO THE LID TOP ALONG BOTH SIDES; AR400: LOW-HYDROGEN ROD, PREHEAT 300 F.

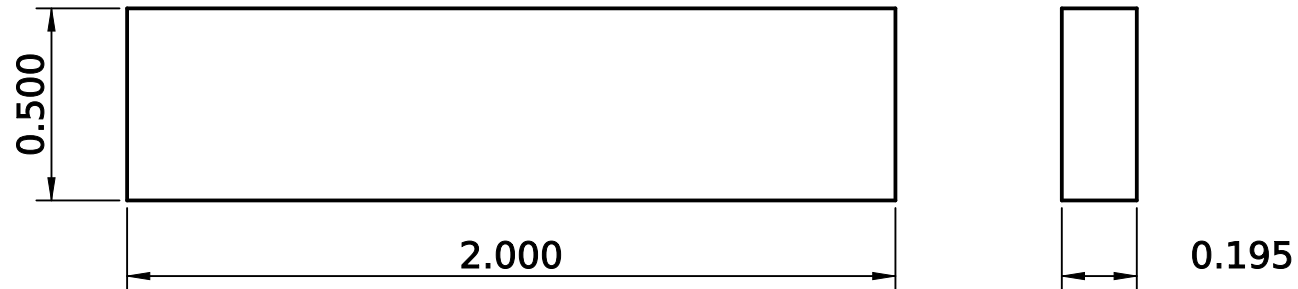
Material 0.750" plate AR400		Qty 2	Scale 1:2	Units inch	Projection
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title V Ramp Simple CINVA-Ram Press, 14x7x4 brick		Identification number SCR-14x7x4-07	Sheet 11/27
				Rev A	Date of issue 2026-09-30



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. OUTSIDE THE SIDE PLATE ON THE LID PIVOT PIN; TOP END WELDED TO THE LID PLATE EDGE.
5. THE LID SWINGS OVER ON THESE AND RESTS UPSIDE DOWN BESIDE THE PRESS.

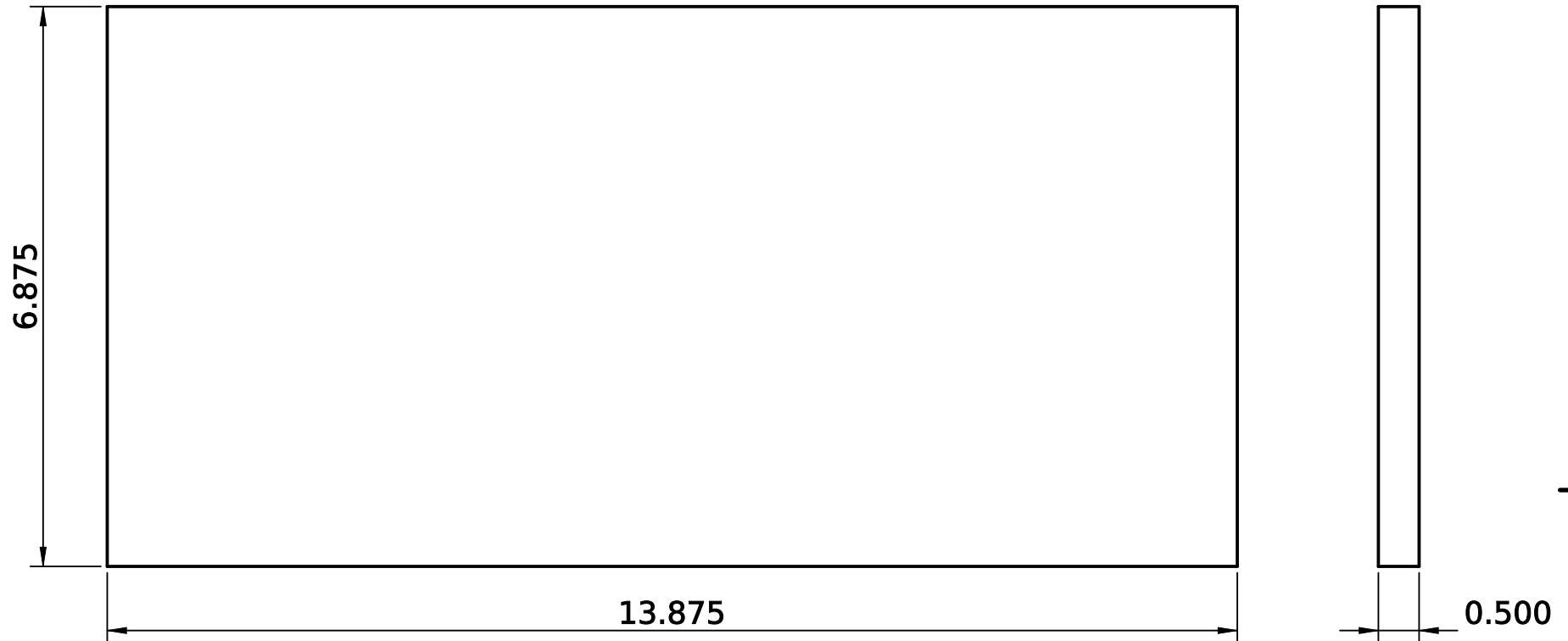
Material 0.375" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Lid Strap Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-08	Sheet 12/27
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD ON THE LID TOP ALONG EACH LONG EDGE AT THE PULL END, UNDER THE HEAD CHEEKS. THE CHEEKS LAND ON THEM 3 DEG PAST CENTRE: THAT IS THE OVER-CENTRE LOCK.
5. FIT ON ASSEMBLY: SHIM OR GRIND SO THE HANDLE STOPS 3 DEG BELOW HORIZONTAL WITH THE LID CLOSED.

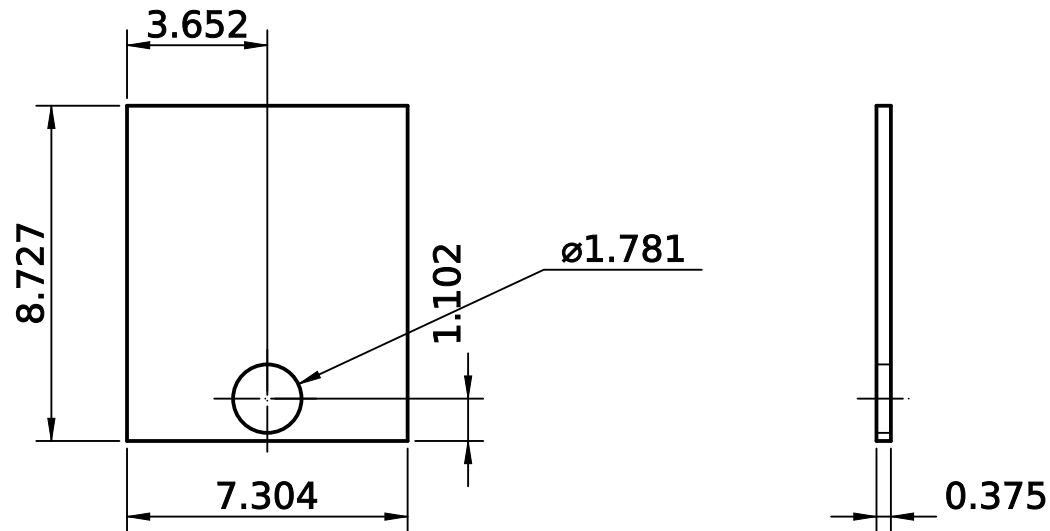
Material 0.195 x 1/2 flat bar		Qty 2	Scale 2:1	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Handle Rest Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-09	Sheet 13/27
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. TOP FACE FLAT; $1/16$ CLEARANCE EACH SIDE IN THE MOLD.

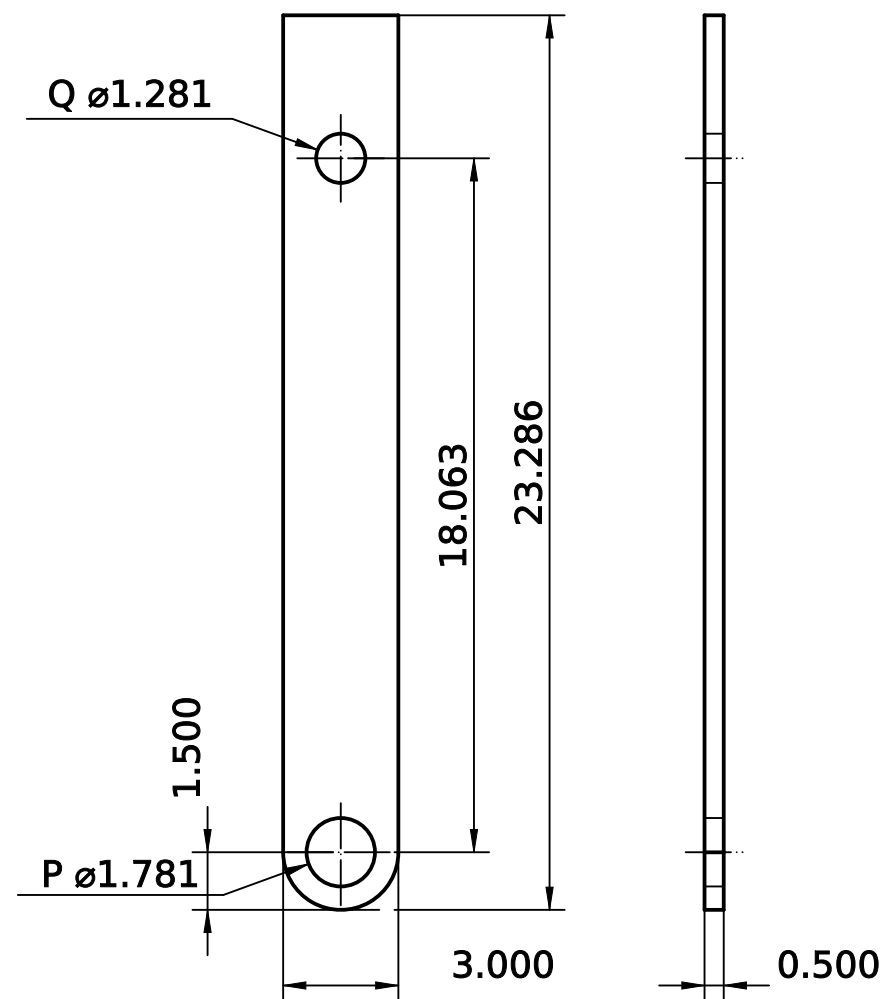
Material 0.500" plate A36		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Piston Cap Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-10		Sheet 14/27
				Rev A	Date of issue 2026-09-30	



NOTES:

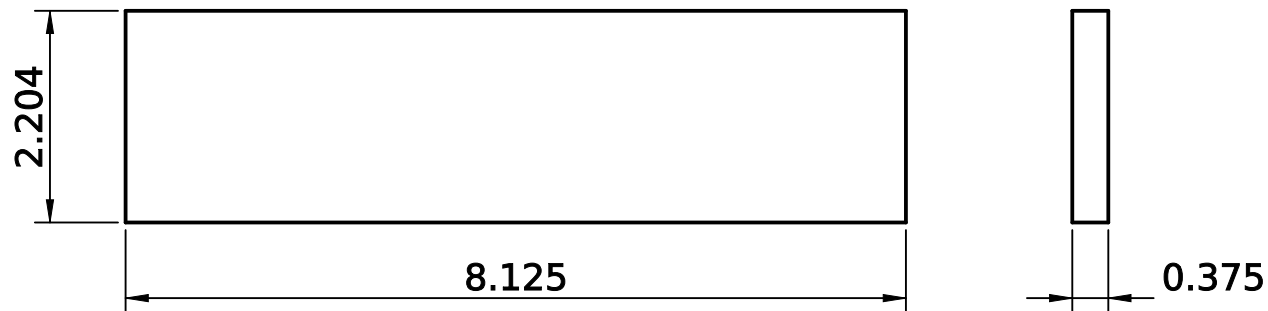
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. TWO PAIRS UNDER THE CAP: OUTER FACES FLUSH WITH THE CAP EDGES, THE OTHERS $1/2$ INBOARD OF THEM. LINE-BORE THE PIN HOLES AFTER WELDING.

Material 0.375" plate A36		Qty 4	Scale 1:5	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Piston Web Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-11		Sheet 15/27
				Rev A	Date of issue 2026-09-30	



- NOTES:
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 2. BREAK SHARP EDGES, REMOVE BURRS.
 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 4. DRILL BOTH ARMS CLAMPED TOGETHER.

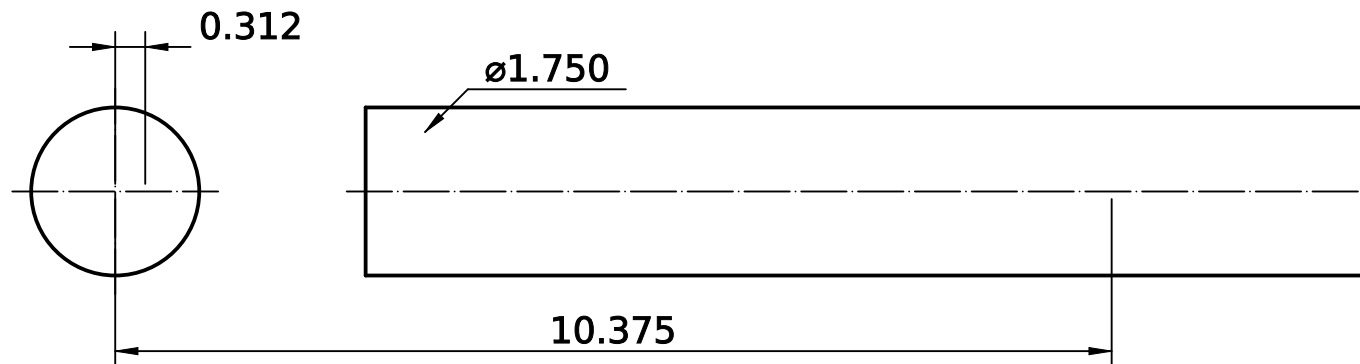
Material 0.500" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Yoke Arm Simple CINVA-Ram Press, 14x7x4 brick		Identification number SCR-14x7x4-12	Sheet 16/27
				Rev A	Date of issue 2026-09-30



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. ON EDGE BETWEEN THE ARM TOPS, ONE FLUSH WITH EACH ARM EDGE (A BOX).
5. THE HEAD RESTS ON THE LOWER EDGE OF THE PULL-END BAR AT THE START, AND THE LATCH HOOKS ITS TOP EDGE.

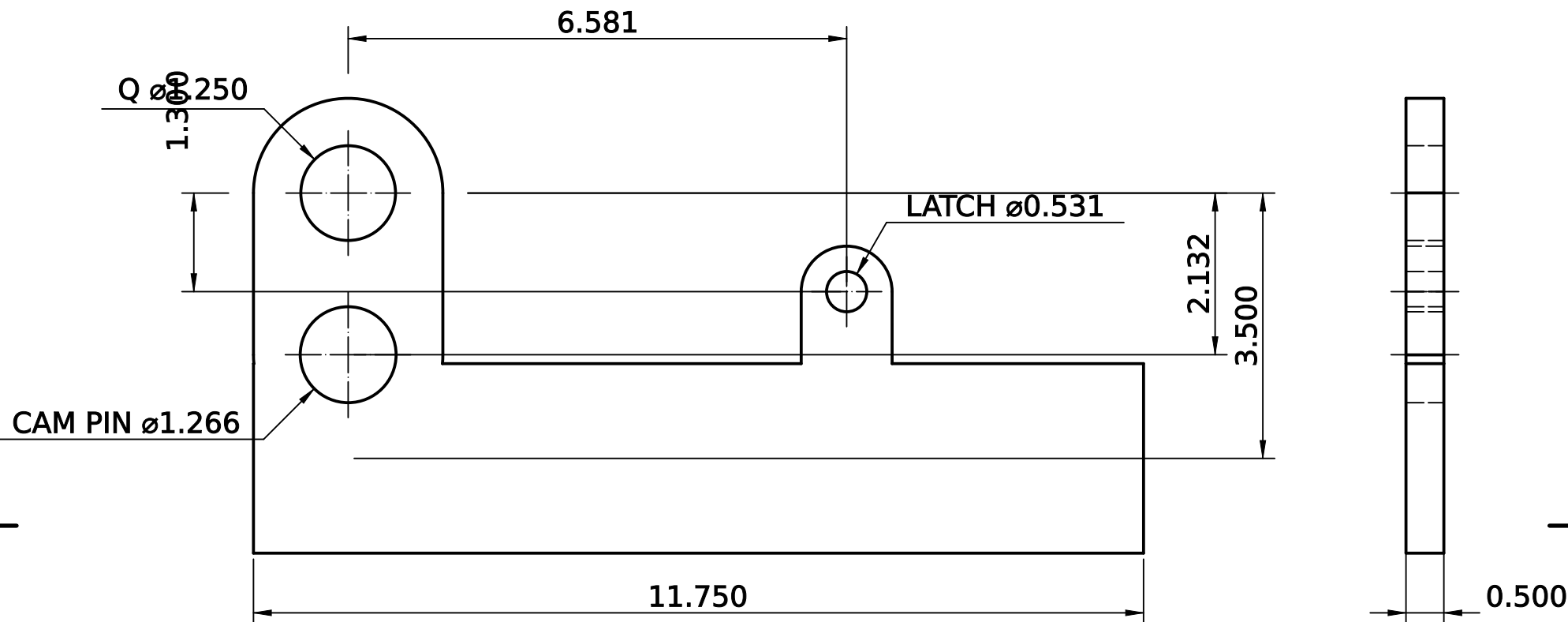
Material 0.375" plate A36		Qty 2	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Cross Bar Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-13	Sheet 17/27
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH ARM, SLOT, FOUR WEBS, SLOT, ARM; $1/4$ HITCH PINS OUTSIDE THE ARMS.

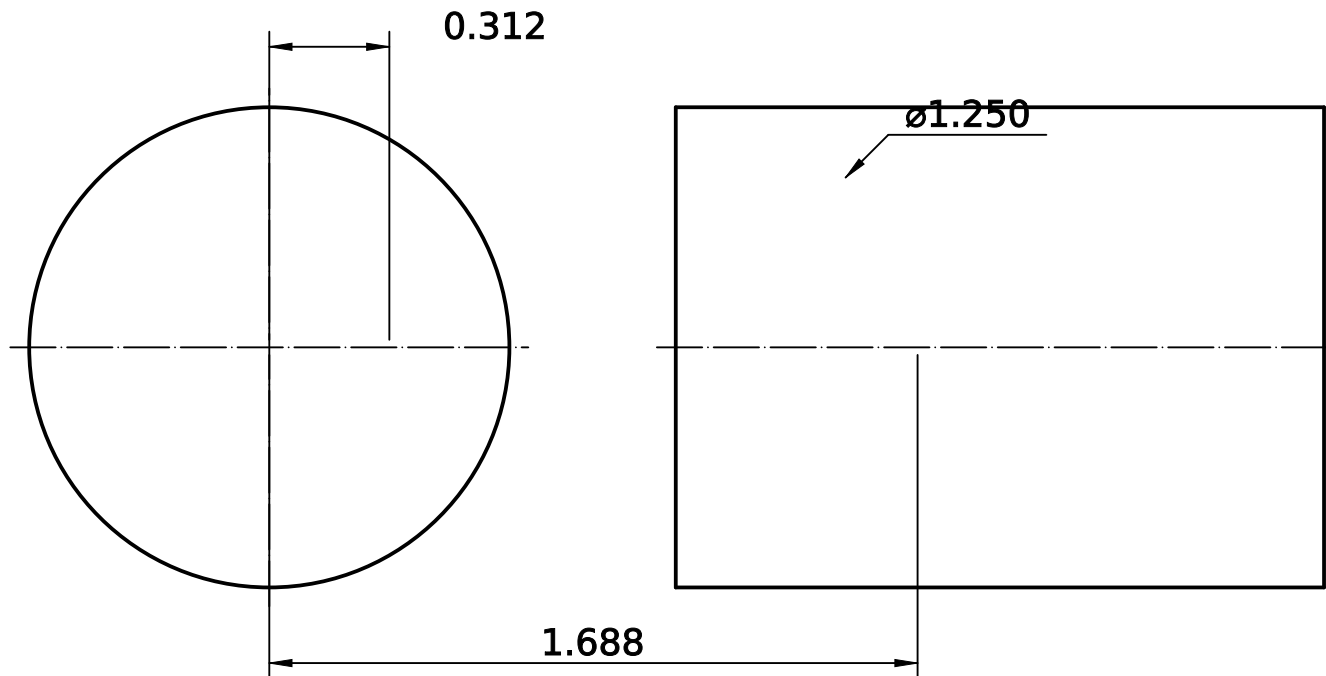
Material 1.750" round 4140 prehard		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -	Document type Detail drawing		Status Draft
Legal owner J. Clements III CEB Project	Title Main Pin P Simple CINVA-Ram Press, 14x7x4 brick	Identification number SCR-14x7x4-14		Sheet 18/27	
		Rev A	Date of issue 2026-09-30		



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THE L-CAM: CAM PIN 2.132 BELOW Q (HOLD ± 0.010), HANDLE LEG ALONG THE OTHER SIDE.
5. HIS HEAD IS BUILT FROM HIS LINK PLATES AND 3 X 2 PLATES; THIS ONE-PIECE CHEEK DOES THE SAME JOB WITH ONE PART PER SIDE.

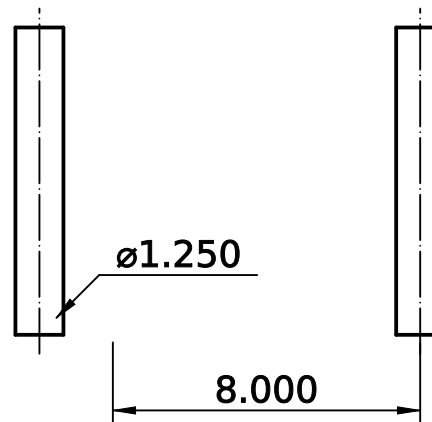
Material 0.500" plate A36	Qty 2	Scale 1:2	Units inch	Projection 
Created by J. Clements III	Approved by -	Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Head Cheek Simple CINVA-Ram Press, 14x7x4 brick	Identification number SCR-14x7x4-15	Sheet 19/27	
		Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD INTO THE CHEEK, INNER END FLUSH INSIDE; TURNS IN THE ARM. WASHER AND HITCH PIN OUTSIDE.

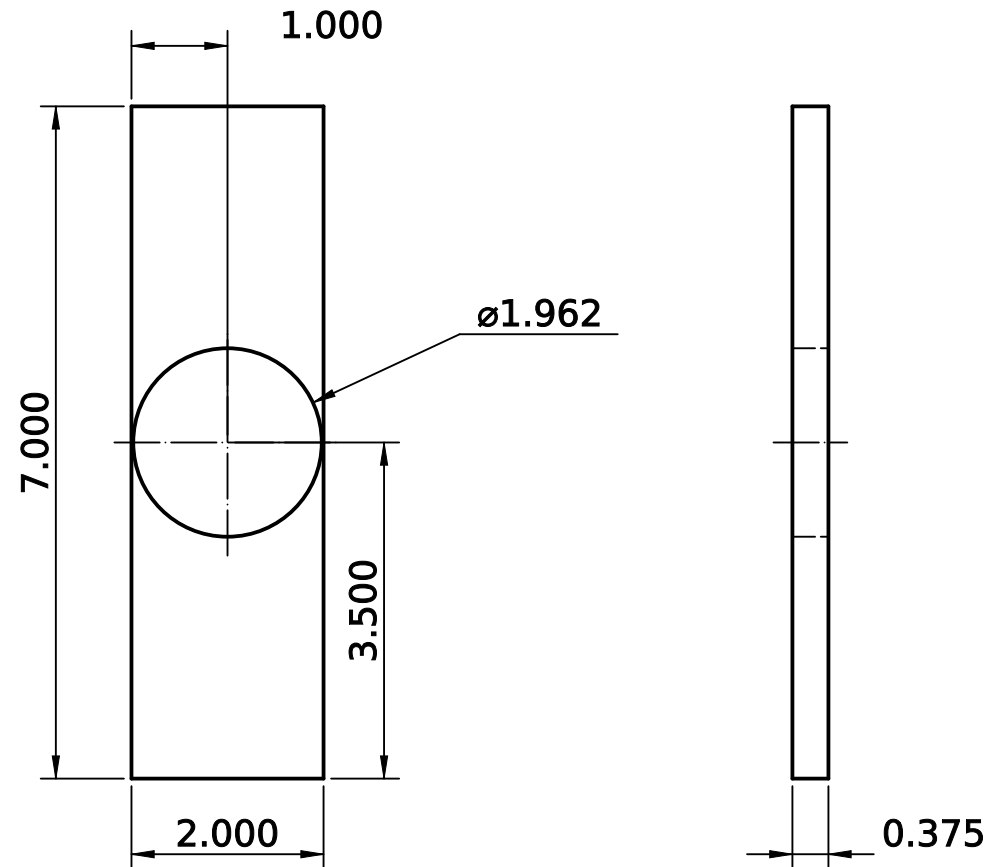
Material 1.250" round 4140 prehard		Qty 2	Scale 2:1	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Pin Q (stub) Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-16		Sheet 20/27
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH CHEEKS, FLUSH OUTSIDE; WELD THE ENDS. IT SEATS IN THE RAMP SCOOPS.

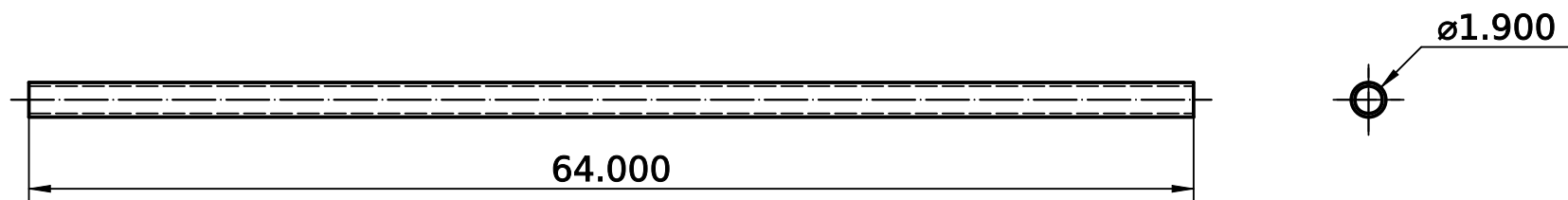
Material 1.250" round 4140 prehard		Qty 1	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Cam Pin Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-17	Sheet 21/27
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BETWEEN THE CHEEKS ON THE HANDLE LEG, SQUARE TO IT, 8 AND 9.75 FROM Q.

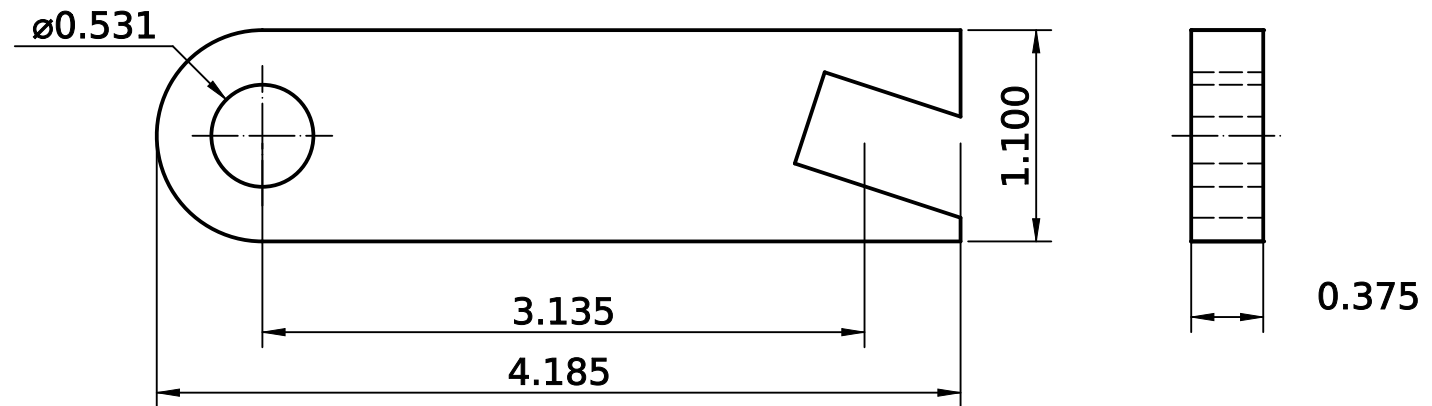
Material 0.375" plate A36		Qty 2	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Handle Bridge Simple CINVA-Ram Press, 14x7x4 brick	Identification number SCR-14x7x4-18			Sheet 22/27
		Rev A	Date of issue 2026-09-30		



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. THROUGH BOTH BRIDGES, END FLUSH WITH THE INNER BRIDGE; WELD ALL ROUND.

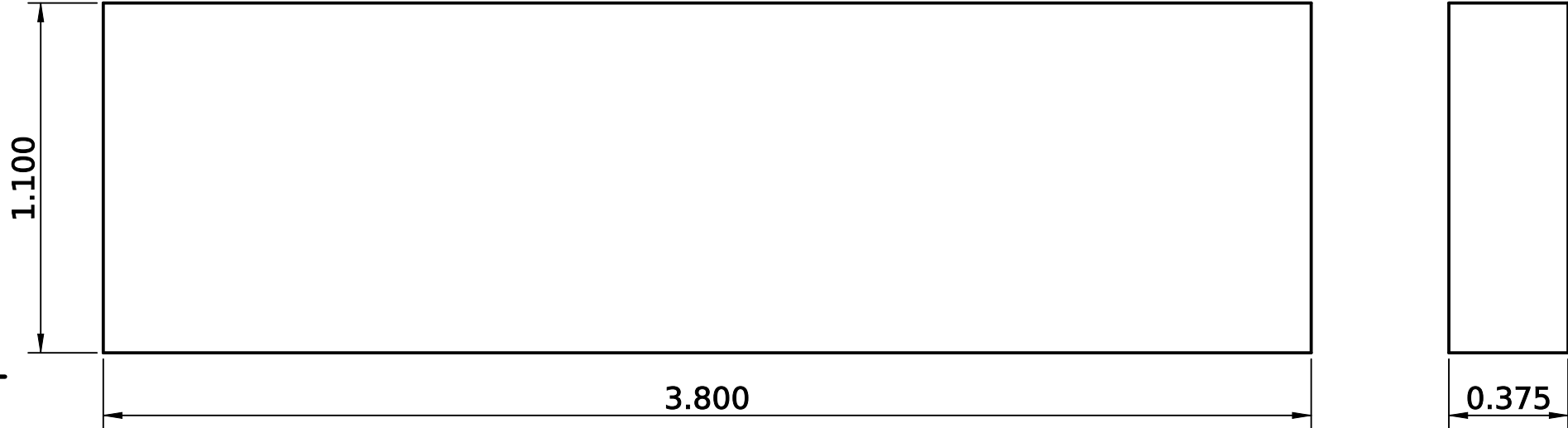
Material 1-1/2" sch 80 pipe		Qty 1	Scale 1:10	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Handle Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-19	Sheet 23/27
				Rev A	



NOTES:

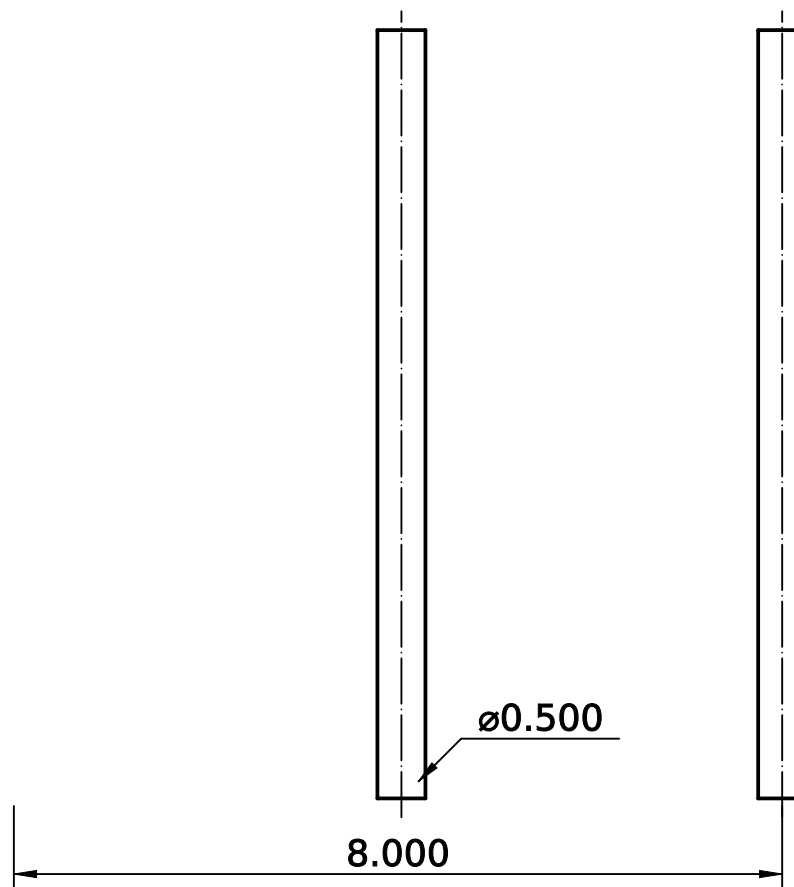
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. FORK AT THE TIP 0.500 WIDE, 0.600 DEEP, FITS OVER THE CROSS BAR.

Material 0.375" plate A36		Qty 2	Scale 1:1	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Latch Hook Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-20		Sheet 24/27
				Rev A	Date of issue 2026-09-30	



- NOTES:
- 1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 - 2. BREAK SHARP EDGES, REMOVE BURRS.
 - 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
 - 4. WELDED ACROSS THE TWO HOOKS; LIFT IT TO OPEN THE LATCH.

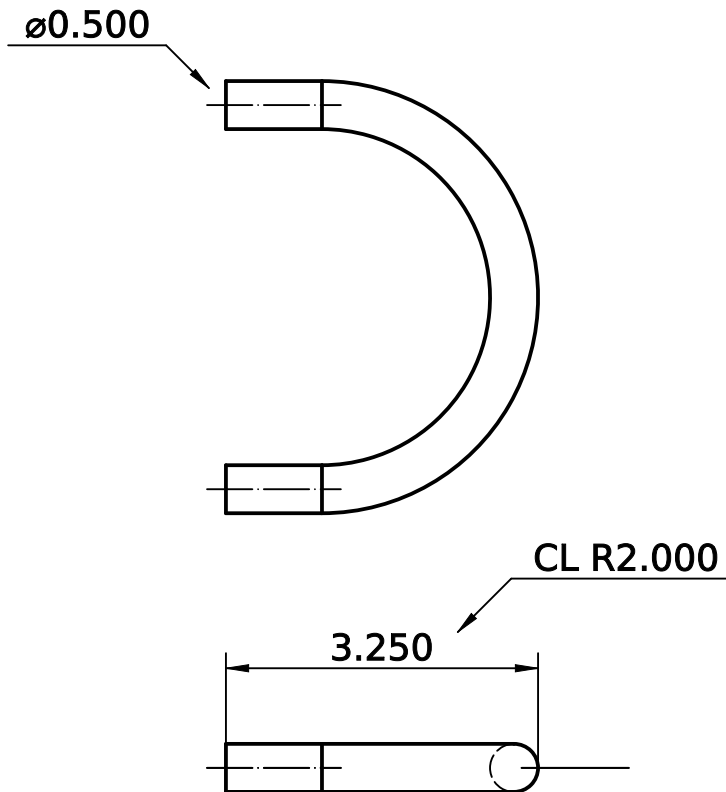
Material 3/8 x 1-1/8 flat bar		Qty 1	Scale 2:1	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Latch Bar Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-21	Sheet 25/27
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH CHEEKS AND HOOKS; COTTER EACH END.

Material 0.500" round 1018		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Latch Pivot Simple CINVA-Ram Press, 14x7x4 brick			Identification number SCR-14x7x4-22		Sheet 26/27
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BEND $1/2$ ROUND INTO A U, 4 BETWEEN LEG CENTRES; WELD THE LEG ENDS TO THE LID'S FIXED-PIN END FACE, CENTRED, LEVEL WITH THE LID.
5. LIFT HERE TO SWING THE LID OVER ONTO THE STAND, AND TO SWING IT BACK.

Material 0.500" round 1018, bent	Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III	Approved by -	Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Lid Handle Simple CINVA-Ram Press, 14x7x4 brick	Identification number SCR-14x7x4-23	Sheet 27/27	
		Rev A	Date of issue 2026-09-30	